Renyi's Entropy

Use the probability in other ways

Shannon's entropy

- Take P and use it as log(1/p)

Generalize

- Take P and use it as f(p)

Renyi's entropy

- Take P and use it as p^{α}, with $\alpha \neq 1$

Various information measures

- Weighted entropy
- Havrda - Charvat entropy
- Tsallis entropy

…

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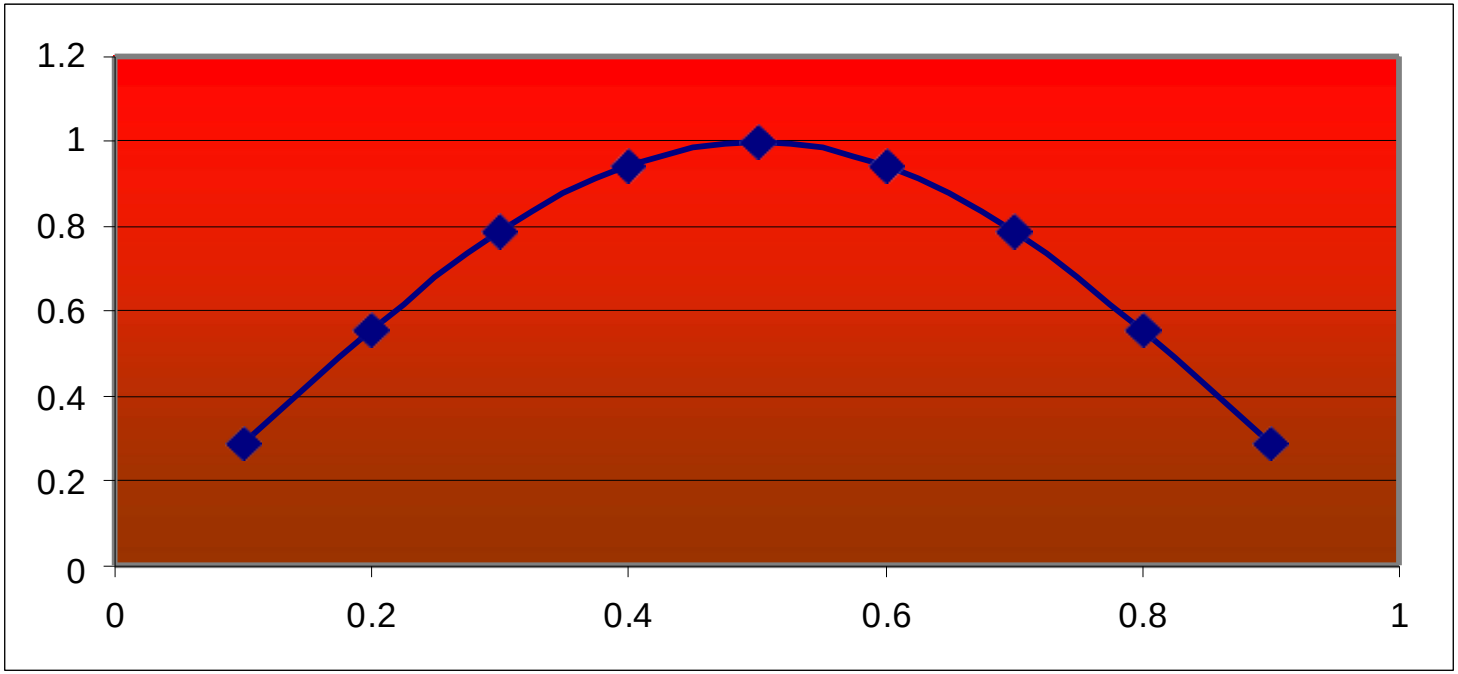
Renyi entropy

- Free parameter: □or q
- Denoted as $H_{R\square}$ or H_{Rq}

$$H_\alpha(X) = \frac{1}{1-\alpha}\log_2\left(\sum_{i=1}^{n} p_i^\alpha\right)$$

H_{R2} versus p (i.e. □=2)

- H_{R2}=-log □$_i$ p_i^2
- Binary outputs with p and (1-p)
- $H_{R2}(\mathbf{X}) = -\log \{p^2 + (1 - p)^2\}$

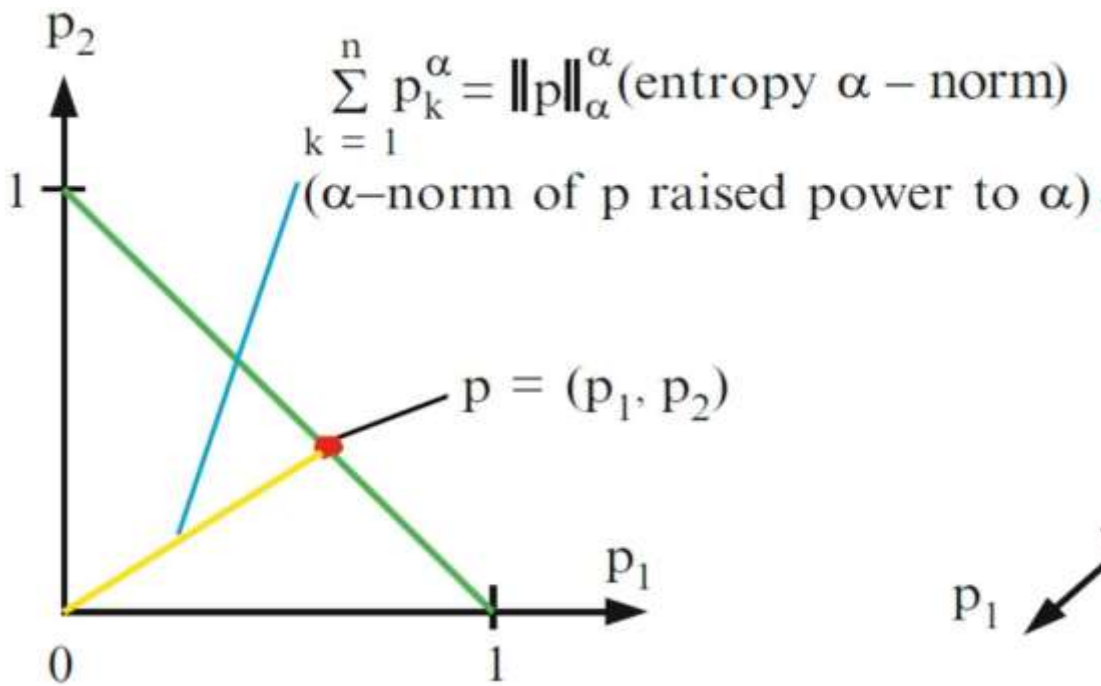


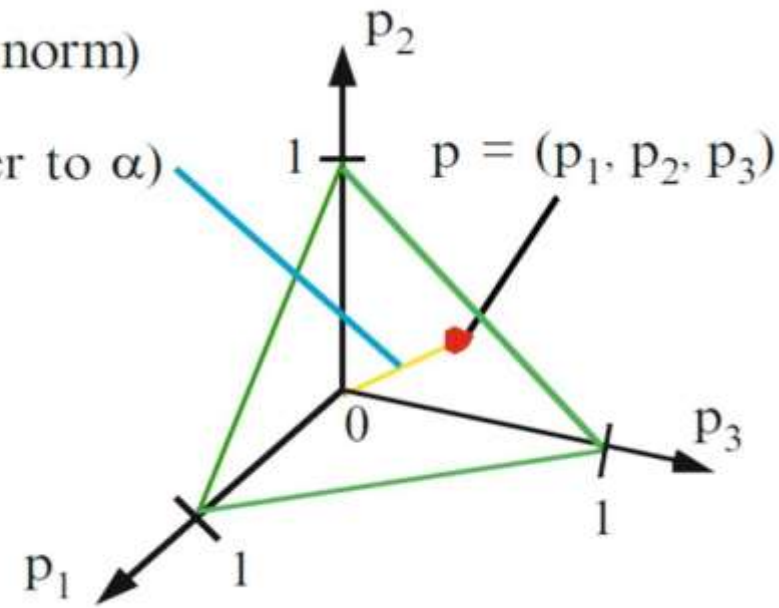
Information potential

$$H_\alpha(X) = \frac{1}{1-\alpha}\log\left(\sum_{k=1}^{N} p_k^\alpha\right) = -\log\left(\sum_{k=1}^{N} p_k^\alpha\right)^{\frac{1}{\alpha-1}}$$

$$H_\alpha(X) = \frac{1}{1-\alpha}\log\left(V_\alpha(X)\right) = -\log\left(\sqrt[\alpha-1]{V_\alpha(X)}\right)$$

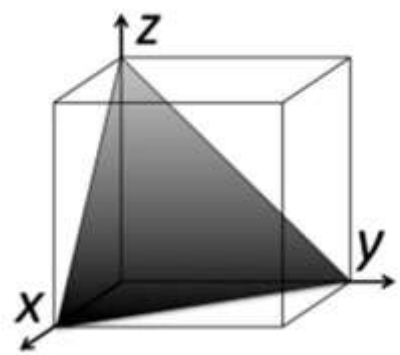
Renyi Entropy – Geometric view





$P_2 = 1 - P_1 = -P_1 + 1$ □m.x + c

Recall p values sum to 1



Renyi entropy in expectation form

$$H_{\alpha} = \frac{1}{1-\alpha} \log \left[\sum_{n} p^{\alpha}(I_n) \right] \quad \text{with } \alpha \neq 1 \text{ and } \alpha > 0$$

$$H_{\alpha} = \frac{1}{1-\alpha} . \log \left[E\left(p^{\alpha-1}\right) \right]$$

General Formula for Rényi Entropy

For a discrete probability distribution $p = (p_1, p_2, \ldots, p_n)$ and order $\alpha > 0, \alpha \neq 1$:

$$H_\alpha(p) = \frac{1}{1-\alpha} \log \left(\sum_{i=1}^{n} p_i^\alpha \right)$$

Expectation Form

Notice that the summation $\sum p_i^\alpha$ can be seen as an **expectation** with respect to the distribution p.

- Write:

$$\sum_{i=1}^{n} p_i^\alpha = \sum_{i=1}^{n} p_i \, p_i^{\alpha-1} = \mathbb{E}_p \left[p^{\alpha-1} \right]$$

So we can rewrite Rényi entropy as:

$$H_\alpha(p) = \frac{1}{1-\alpha} \log \mathbb{E}_p \left[p^{\alpha-1} \right]$$

Slide Breakdown

- The slide is showing this equivalence:

1. **Summation Form:**

$$H_\alpha = \frac{1}{1-\alpha} \log \left(\sum_{i=1}^{n} p_i^\alpha \right)$$

2. **Expectation Form:**

2. **Expectation Form:**

$$H_\alpha = \frac{1}{1-\alpha} \log \left(\mathbb{E}_p[p^{\alpha-1}] \right)$$

- Here, $\mathbb{E}_p[\cdot]$ means expectation with respect to the distribution p.
- Inside, $p^{\alpha-1}$ means each probability p_i raised to power $\alpha - 1$.

Conditions

- The slide also notes:
  $\alpha \neq 1$ (because for $\alpha = 1$, this formula collapses and gives **Shannon entropy** instead).
  $\alpha > 0$ (to ensure well-definedness of the power terms).

✅ **Summary:**

This slide reformulates Rényi entropy from a **sum of powers of probabilities** into an **expectation form**. This is elegant because it highlights that Rényi entropy can be interpreted as the logarithm of the expected value of $p^{\alpha-1}$, normalized by $\frac{1}{1-\alpha}$.

1. Recap: Rényi Entropy

The formula at the top:

$$H_\alpha(X) = \frac{1}{1-\alpha}\log_2\left(\sum_{i=1}^{n} p_i^\alpha\right)$$

- This is the Rényi entropy of order α.
- When $\alpha \to 1$, it reduces to **Shannon entropy**:

$$H(X) = -\sum p_i \log p_i$$

2. Rényi Divergence

Rényi divergence compares two probability distributions $P = (p_1, \ldots, p_n)$ and $Q = (q_1, \ldots, q_n)$.

The formula:

$$D_\alpha(P\|Q) = \frac{1}{\alpha-1}\log\left(\mathbb{E}_P\left[\left(\frac{p}{q}\right)^{\alpha-1}\right]\right)$$

Breaking it down:

- **Expectation under P** means:

$$\mathbb{E}_P\left[f\left(\frac{p}{q}\right)\right] = \sum_i p_i f\left(\frac{p_i}{q_i}\right)$$

- The function here is $\left(\frac{p}{q}\right)^{\alpha-1}$.

So:

Renyi divergence

$$H_\alpha(X) = \frac{1}{1-\alpha}\log_2\left(\sum_{i=1}^{n} p_i^\alpha\right)$$

Renyi entropy

$$H = \frac{1}{\alpha \quad 1} log\left(E\left(p^{(\alpha-1)}\right)\right)$$

Renyi divergence

$$D(P||Q) = \frac{1}{\alpha-1} log\left(E\left(\frac{p}{q}\right)^{(\alpha-1)}\right)$$

Weighing factor is p

$$D(P||Q) = \frac{1}{\alpha-1} log \sum \frac{p^{(\alpha)}}{q^{(\alpha-1)}}$$

With □=1, Renyi divergence=KLD

- The function here is $\left(\frac{p}{q}\right)$.

So:

$$D_\alpha(P\|Q) = \frac{1}{\alpha-1}\log\left(\sum_i p_i^\alpha q_i^{1-\alpha}\right)$$

3. Connection to KL Divergence

- When $\alpha \to 1$, Rényi divergence converges to the **Kullback–Leibler divergence (KLD)**:

$$D(P\|Q) = \sum_i p_i \log\frac{p_i}{q_i}$$

That's why the slide notes:
With $\alpha \to 1$, Rényi divergence = KLD.

4. Summary of Slide

- **Rényi entropy** is about the "spread" of one distribution.
- **Rényi divergence** extends this to comparing **two distributions**.
- Formula:

$$D_\alpha(P\|Q) = \frac{1}{\alpha-1}\log\sum_i \frac{p_i^\alpha}{q_i^{\alpha-1}}$$

- For $\alpha = 1$, it reduces to **KL divergence**.